Computational studies of uniform and boundary-frustrated quantum spin chains

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Abstract

Phase transitions are very interesting phenomena in various branches of physics and always attract a lot of researchers from different field because they appear around us in real life on a daily basis. In this work, we study the quantum phase transition by solving the transverse field Ising model analytically, then compare the solution with the computational simulation when changing the transverse field. Following that, we keep using the transverse field Ising model to observe another quantum phase transition which is motivated by geometrical frustration and driven by a defected bond interaction. We discuss about the phase transitions and investigate under which circumstances the system is frustrated.

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Chapter 1

Introduction

Phase transitions are an interesting phenomena when a substance 'suddenly' changes its phase. A phase transition does not occur instantaneously, it is an abrupt change of the properties as a function of the system parameters. One of the most common examples of phase transitions are the freezing and boiling of water. These transitions are associated with an abrupt change in the density in the case of boiling, and a sudden emergence of order in the case of freezing. These fascinating phenomena have been poorly understood for a long time. It seems paradoxical that although we believe the world around us to be governed by smooth functions and differential equations leading to analytical solutions, we see such abrupt, non-analytic behaviour. In the first half of the twentieth century it has become clear that the reason for this non-analytic behaviour lies in the fact that macroscopic objects (such as a glass of water) consist of large (almost infinite) numbers of molecules: a function depending on a parameter, which is analytic for every finite value of that parameter, may become non-analytic if the parameter becomes infinite. In our case, this parameter is the number of particles (or, more generally, the number of degrees of freedom). A phase transition is always characterised by a sudden change in the degree or the type of order in the system. In order to analyse phase transitions, it is necessary to always identify a parameter which characterises the degree and/or type of order present in the system. This parameter is called the order parameter. It can be a scalar, a vector, a tensor, a complex number, etc. An order parameter is zero in the disordered phase (above the critical point) and non-zero in the ordered phase (below the critical point), and is usually predicted in phenomenological theory and experimental results first, and then confirmed by theoretical calculations. Landau theory [1] is a famous phenomenological theory to predict what happens at and near the critical point by expanding the free energy as a power series in the order parameter. Conventionally, the phase transitions studied are driven by temperature, and have a critical temperature associated with them. However, all such phenomena need not be temperature driven. There is another type of phase transition, so-called quantum phase transition, that is driven by quantum instead of thermal fluctuations. Just like the 2D classical Ising model is usually used to observe the thermal phase transitions, we use the transverse field Ising model as a prototypical example of quantum phase transition. In the next part, another interesting quantum phase transition motivated

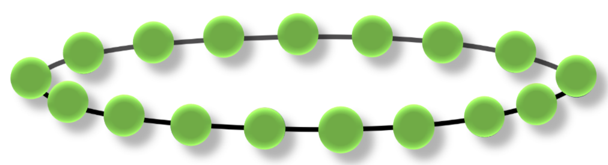


Figure 1.1: A schematic digram of a spin chain.

by frustration is also studied in the transverse field Ising model. Frustration is the consequence of the competition between interacting components in the Ising ring, so that all the interactions of the system cannot be minimized simultaneously. We investigate this phenomenon by a bond defect in the transverse field Ising model.

Chapter 2

Transverse Field Ising model

The one dimensional (1D) Quantum Transverse Field Ising model will be introduced in this chapter. It it the prototypical and simplest non-trivial quantum many-body system to study and observe the quantum phase transition.

2.1 General Ising model

Ising model (or also known as Lenz-Ising model) was firstly suggested to Ernst Ising by his advisor, Wilhelm Lenz (1920) as a realistic representation of a ferromagnet [2]. Ising then exactly solved that model in one-dimensional space in his PhD dissertation (1924) and proved that magnetic phase transition does not exist in one dimension [3]. Let us write down the formulation of the original model. In Ising's dissertation, he only describes the model verbally and argumentatively calculates the partition function. The corresponding Hamiltonian, however, can be written as:

$$\widehat{H}_{IM} = -J_{ij} \sum_{\langle i,j \rangle} \hat{\sigma}_i^x \hat{\sigma}_j^x, \tag{2.1}$$

where $\hat{\sigma}_i^x$ is the Pauli matrix in the preferred x-direction (Appendix A), and $J_{ij} > 0$ are the ferromagnetic interactions. i and j denote the positions of spins in the one-dimensional chain, and $\langle i, j \rangle$ means that all pairs of nearest-neighbor spins are included into the sum.

The assumptions of the model are that the elementary magnets only have two possibilities: up $(+)$ and down $(-)$, and only the interactions between two very neighboring spins are considered. Another common simplification is to assume that all of the nearest neighbors $\langle i, j \rangle$ have the same interaction strength. So we can rewrite the equation 2.1 as

$$\widehat{H}_{IM} = -J \sum_j \hat{\sigma}_j^x \hat{\sigma}_{j+1}^x. \tag{2.2}$$

2.2 Transverse Field Ising model (TFIM)

The original model aims to examine the spontaneous magnetization at low temperature. However, in order to study the quantum phase transitions, we want to add another term that is transverse magnetic field with strength h:

$$\widehat{H}_{TF} = -h \sum_j \hat{\sigma}_j^z. \tag{2.3}$$

This new field introduces the quantum fluctuations, which is very similar to the thermal fluctuations introduced by the temperature in the classical Ising system.
By summing the original Ising and transverse-field parts, we have the final form of the transverse field Ising model (TFIM) Hamiltonian [4]:

$$\widehat{H}_{TFIM} = \widehat{H}_{IM} + \widehat{H}_{TF} = -J \sum_j \hat{\sigma}_j^x \hat{\sigma}_{j+1}^x - h \sum_j \hat{\sigma}_j^z. \tag{2.4}$$

The minus sign on each term of the Hamiltonian is just conventional. According to this sign convention of J, TFIM is able to possess two different types of interactions:

- Ferromagnetic interaction: $J > 0$,
- Antiferromagnetic interaction: $J < 0$.

2.3 Spontaneous symmetry breaking (SSB) in 1D TFIM

As we mentioned in the last section, in order to see the quantum phase transition, we choose to study the 1D TFIM Hamiltonian at zero temperature like in the expression

(2.4). For this purpose, let us introduce the parity operator:

$$\widehat{P} = \prod_j \hat{\sigma}_j^z. \tag{2.5}$$

Basically, this operator flips all the spins in x-axis and just counts the parity of the spins in z-axis, like the following:

$$\hat{\sigma}_i^x \rightarrow \widehat{P}^\dagger \hat{\sigma}_i^x \widehat{P} = -\hat{\sigma}_i^x, \tag{2.6}$$

and

$$\hat{\sigma}_i^z \rightarrow \widehat{P}^\dagger \hat{\sigma}_i^z \widehat{P} = \hat{\sigma}_i^z. \tag{2.7}$$

Hence the parity operator commutes with TFIM Hamiltonian $\widehat{H}_{TFIM}$, so it generates a global $\mathbb{Z}_2$ symmetry of the Hamiltonian. Even though there is no thermal fluctuation at zero temperature, quantum fluctuations are introduced by the term $-h\sum_j \hat{\sigma}_j^z$, which flips the spins and destroy the order. In the next few sections, the exact solution of TFIM will be presented, but let us first consider two trivial limiting cases of the model.

2.3.1 Strong field or weak interaction limit ($h >> J$ or $J \rightarrow 0$)

In this limit, the Ising exchange interactions between spins are vanishingly small, only the transverse magnetic field term is relevant. All the spins want to align with the field (point parallel to the magnetic field along the z-axis). There is no spontaneously broken symmetry, hence this state is quantum-mechanically unordered, also called paramagnetic state. The ground state of the system is unique in this case:

$$|0_h\rangle = \prod_j |\uparrow\rangle_j = |\uparrow\uparrow\uparrow \ldots \uparrow\uparrow\uparrow\rangle\,, \tag{2.8}$$

and the corresponding ground-state energy is given as:

$$E_0^h = \langle 0_h| \lim_{h\rightarrow\infty} \widehat{H}_{TFIM} \,|0_h\rangle = -Lh, \tag{2.9}$$

where L is the number of sites in the chain.

2.3.2 Weak field or strong interaction limit ($h << J$ or $h \to 0$)

In this limit, the transverse-field term is vanishingly small, the original Ising model is back. The system prefers to have a state in which all the nearest neighbor spins align in the same x-direction, but there are actually two possibilities of ground state which carry the same minimum energy now:

$$\left|0_J^r\right\rangle = \prod_j \left|\rightarrow\right\rangle_j = \left|\rightarrow\rightarrow\rightarrow \ldots \rightarrow\rightarrow\rightarrow\right\rangle, \tag{2.10}$$

and

$$\left|0_J^l\right\rangle = \prod_j \left|\leftarrow\right\rangle_j = \left|\leftarrow\leftarrow\leftarrow \ldots \leftarrow\leftarrow\leftarrow\right\rangle. \tag{2.11}$$

with the ground state energy

$$E_0^{J>0} = \left\langle 0_J^r\right| \lim_{J\to\infty} \widehat{H}_{TFIM} \left|0_J^r\right\rangle = \left\langle 0_J^l\right| \lim_{J\to\infty} \widehat{H}_{TFIM} \left|0_J^l\right\rangle = -LJ. \tag{2.12}$$

However, we should note that the Hamiltonian always wants to retain the $\mathbb{Z}_2$ symmetry which means it does not distinguish between $\left|0_J^r\right\rangle$ and $\left|0_J^l\right\rangle$, so the ground state if we solve the Hamiltonian by diagonalization will be the symmetric combination of $\left|0_J^r\right\rangle$ and $\left|0_J^l\right\rangle$:

$$\left|0_J^\pm\right\rangle = \frac{\left|0_J^r\right\rangle \pm \left|0_J^l\right\rangle}{\sqrt{2}}. \tag{2.13}$$

An interesting point is that if we change the sign of J (i.e., $J < 0$), the ground state will be different. The neighboring spins want to be aligned antiparallel (Néel state) instead of parallel like when $J > 0$:

$$\left|0_J^{rl}\right\rangle = \prod_j \left|\rightarrow\right\rangle_j = \left|\rightarrow\leftarrow\rightarrow \ldots \rightarrow\leftarrow\rightarrow\right\rangle, \tag{2.14}$$

and

$$\left|0_J^{lr}\right\rangle = \prod_j \left|\leftarrow\right\rangle_j = \left|\leftarrow\rightarrow\leftarrow \ldots \leftarrow\rightarrow\leftarrow\right\rangle. \tag{2.15}$$

But the state still has the same energy as the ferromagnetic state:

$$E_0^{J<0} = \left\langle 0_J^{rl} \right| \lim_{J\to\infty} \widehat{H}_{TFIM} \left| 0_J^{rl} \right\rangle = \left\langle 0_J^{lr} \right| \lim_{J\to\infty} \widehat{H}_{TFIM} \left| 0_J^{lr} \right\rangle = LJ. \tag{2.16}$$

2.3.3 Spontaneous Symmetry Breaking

After considering two different limiting cases, we have found that the weak and strong field limits of TFIM Hamiltonian behave qualitatively differently. Note that the $\mathbb{Z}_2$ generator, $\widehat{P} = \prod_j \hat{\sigma}_j^z$, which commutes with $\widehat{H}_{TFIM}$, transforms the states $|\rightarrow\rangle$ and $|\leftarrow\rangle$ into one another. When the external field h is very large, the TFIM ground state is still parity symmetric as $\widehat{P}$ does not transform the state. However, with zero field case or when the interaction is extremely large compared to the external field, there is two-fold degenerate state (even with positive or negative J). When entering the phase from the paramagnetic phase, the system needs to decide for one of the two configurations and the symmetry is spontaneously broken. In the next section, we will show the exact solution of the model.

2.4 Analytical solution of 1D TFIM

2.4.1 Jordan-Wigner transformation

The main idea behind the exact solution of the transverse-field Ising model is the realization that spin-1/2 degrees of freedom are very similar to spinless fermions, through the following assocation:

Spins	**Fermions**
spin down	occupied site
spin up	empty site

or in terms of operators:

$$\hat{\sigma}_j^z = 1 - 2\hat{c}_j^\dagger \hat{c}_j, \tag{2.17}$$

where $\hat{c}_j^\dagger$ and $\hat{c}_j$ are creation and annihilation operators of fermion systems, respectively. We are tempted to complete the mapping by associating $\hat{\sigma}_j^+ = (\hat{\sigma}_j^x + i\hat{\sigma}_j^y)/2$ with $\hat{c}_j$ and $\hat{\sigma}_j^- = (\hat{\sigma}_j^x - i\hat{\sigma}_j^y)/2$ with $\hat{c}_j^\dagger$. This gives the correct spin algebra on a single site, but when multiple sites are present, it leads to a problem that $\hat{\sigma}_j^\pm$ commute on different sites whereas $\hat{c}_j^+$ and $\hat{c}_i$ anticommute each other for $j \neq i$ too.

Jordan and Wigner [3] provided the solution that the naive single-site correspondence must be modified by a "string" operator,

$$\hat{\sigma}_j^+ = \left[\prod_{i<j}(1 - 2\hat{c}_i^\dagger\hat{c}_i)\right]\hat{c}_j, \tag{2.18}$$

and

$$\hat{\sigma}_j^- = \left[\prod_{i<j}(1 - 2\hat{c}_i^\dagger\hat{c}_i)\right]\hat{c}_j^\dagger. \tag{2.19}$$

The string $\prod_{i<j}(1 - 2\hat{c}_i^\dagger\hat{c}_i)$ takes on values ± 1, depending on whether an even/odd number of fermions is present to the left of site j. The string solves the statistics problem. For example, let's consider (without loss of generality) that $l > j$, then

$$\hat{\sigma}_j^+\hat{\sigma}_l^- = \left[\prod_{i<j}(1 - 2\hat{c}_i^\dagger\hat{c}_i)\right]\hat{c}_j\left[\prod_{m<l}(1 - 2\hat{c}_m^\dagger\hat{c}_m)\right]\hat{c}_l^\dagger. \tag{2.20}$$

If we move $\hat{c}_j$ all the way to the right, we will pick up the two minus signs, one from switching the parity of the l-string and the second from anticommuting with $\hat{c}_l^\dagger$. And note that the j-string commutes with everything else and we can move it at will. Hence,

$$\hat{\sigma}_j^+\hat{\sigma}_l^- = \left[\prod_{m<l}(1 - 2\hat{c}_m^\dagger\hat{c}_m)\right]\hat{c}_l^\dagger\left[\prod_{i<j}(1 - 2\hat{c}_i^\dagger\hat{c}_i)\right]\hat{c}_j. \tag{2.21}$$

Thus we have successed in mapping spins into spinless fermions. The price we paid is using a highly non-local representation in terms of fermionic operators. We can also write the inverse relations,

$$\hat{c}_j = \left(\prod_{i<j}\hat{\sigma}_i^z\right)\hat{\sigma}_j^+, \tag{2.22}$$

$$\hat{c}_j^\dagger = \left(\prod_{i<j} \hat{\sigma}_i^z\right) \hat{\sigma}_j^-. \tag{2.23}$$

It is easy to check that the following relations are correct:

$$\{\hat{c}_j, \hat{c}_i^\dagger\} = \delta_{ij} \tag{2.24}$$

and

$$\{\hat{c}_j, \hat{c}_i\} = \{\hat{c}_j^\dagger, \hat{c}_i^\dagger\} = 0. \tag{2.25}$$

which imply that

$$\left[\hat{\sigma}_j^+, \hat{\sigma}_i^-\right] = \delta_{ij}\hat{\sigma}_j^z \tag{2.26}$$

and

$$\left[\hat{\sigma}_j^z, \hat{\sigma}_i^\pm\right] = \pm 2\delta_{ij}\hat{\sigma}_j^\pm \tag{2.27}$$

and vice versa.

2.4.2 The exact solution of nearest neighbor 1D TFIM

The standard form of the Jordan-Wigner transformation was presented. Now we want to insert that into $\widehat{H}_{TFIM}$. Recall the mappings:

$$\hat{\sigma}_j^z = 1 - 2\hat{c}_j^\dagger \hat{c}_j, \tag{2.28}$$

$$\hat{\sigma}_j^x = \prod_{i<j}(1 - 2\hat{c}_j^\dagger \hat{c}_j)(\hat{c}_j + \hat{c}_j^\dagger). \tag{2.29}$$

We can see that the term $\hat{\sigma}_j^z$ of Hamiltonian is very straight forward, let's see $\hat{\sigma}_j^x \hat{\sigma}_{j+1}^x$ term:

$$\begin{aligned}
\hat{\sigma}_j^x \hat{\sigma}_{j+1}^x &= \left[\prod_{i<j}(1 - 2\hat{c}_j^\dagger \hat{c}_j)\right](\hat{c}_j + \hat{c}_j^\dagger)\left[\prod_{i<j+1}(1 - 2\hat{c}_j^\dagger \hat{c}_j)\right](\hat{c}_{j+1} + \hat{c}_{j+1}^\dagger) \\
&= (\hat{c}_j + \hat{c}_j^\dagger)(1 - 2\hat{c}_j^\dagger \hat{c}_j)(\hat{c}_{j+1} + \hat{c}_{j+1}^\dagger) \\
&= (-\hat{c}_j + \hat{c}_j^\dagger)(\hat{c}_{j+1} + \hat{c}_{j+1}^\dagger) \\
&= \hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_{j+1}^\dagger \hat{c}_j + \hat{c}_j^\dagger \hat{c}_{j+1}^\dagger + \hat{c}_{j+1}\hat{c}_j.
\end{aligned} \tag{2.30}$$

The is one important point we should notice when we have to deal with the boundary conditions. If we uses the open boundary conditions, the sum in the interaction term runs over $j = 1, ..., L-1$ which means there is no term involving site $L+1 \equiv 1$, hence we get:

$$\widehat{H}_{\mathrm{OBC}} = -J \sum_{j=1}^{L-1} \left(\hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_j^\dagger \hat{c}_{j+1}^\dagger + \text{ H.c. } \right) + h \sum_{j=1}^{L} \left(2\hat{c}_j^\dagger \hat{c}_j - 1 \right). \tag{2.31}$$

It is a bit different for the periodic boundary conditions which come from:

$$\begin{aligned} \hat{\sigma}_L^x \hat{\sigma}_1^x = \hat{\sigma}_L^x \hat{\sigma}_{L+1}^x &= e^{i\pi \sum_{j=1}^{L-1} \hat{n}_j} \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c.} \right) \\ &= -e^{i\pi \sum_{j=1}^{L} \hat{n}_j} \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c.} \right) \\ &= -e^{i\pi \hat{N}} \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c.} \right) \\ &= -(-1)^{\hat{N}} \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c.} \right) \end{aligned} \tag{2.32}$$

where

$$\widehat{N} = \sum_{j=1}^{L} \hat{c}_j^\dagger \hat{c}_j, \tag{2.33}$$

is the total number of particles. Notice that the second line equality follows because to the left of $\hat{c}_L^+$ we are sure to have $\hat{n}_L = 1$, so that $e^{i\pi \hat{n}_L} = -1$. We see that the Hamiltonian in periodic boundary conditions is different from the one in open boundary conditions by the new term as above, with the sign depends on the total number of particles. We will focus on this boundary condition from now on.

$$\widehat{H}_{PBC} = \widehat{H}_{OBC} + (-1)^{\hat{N}} J \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c. } \right) + h \sum_{j=1}^{L} \left(2\hat{c}_j^\dagger \hat{c}_j - 1 \right). \tag{2.34}$$

If we define the projectors on the even and odd number of particles subspaces:

$$\widehat{\mathrm{P}}_{\mathrm{odd}} = \frac{1}{2}(\hat{1} + e^{i\pi \widehat{N}}) = \widehat{\mathrm{P}}_0, \tag{2.35}$$

and

$$\widehat{\mathrm{P}}_{\mathrm{even}} = \frac{1}{2}(\hat{1} - e^{i\pi \widehat{N}}) = \widehat{\mathrm{P}}_1. \tag{2.36}$$

Then two spinless fermionic Hamiltonians acting on the 2^{L-1}-dimensional even/odd parity subspaces of the full Hilbert space can be defined as:

$$\widehat{\mathrm{H}}_0 = \widehat{\mathrm{P}}_0 \widehat{H}_{PBC} \widehat{\mathrm{P}}_0, \tag{2.37}$$

and

$$\widehat{\mathrm{H}}_1 = \widehat{\mathrm{P}}_1 \widehat{H}_{PBC} \widehat{\mathrm{P}}_1. \tag{2.38}$$

By using the above terms, the full spinless fermionic Hamiltonian can be expressed in block form as:

$$\widehat{H}_{\mathrm{PBC}} = \left(\begin{array}{c|c} \widehat{\mathrm{H}}_0 & 0 \\ \hline 0 & \widehat{\mathrm{H}}_1 \end{array} \right). \tag{2.39}$$

It can be realized that if we write the fermionic Hamiltonian of the form:

$$\begin{aligned} \widehat{\mathbb{H}}_{\mathrm{p}=0,1} = &- J \sum_{j=1}^{L-1} \left(\hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_j^\dagger \hat{c}_{j+1}^\dagger + \text{ H.c. } \right) + (-1)^{\mathrm{p}} J \left(\hat{c}_L^\dagger \hat{c}_1 + \hat{c}_L^\dagger \hat{c}_1^\dagger + \text{ H.c. } \right) \\ &+ \sum_{j=1}^{L} h_j \left(2\hat{n}_j - 1 \right), \end{aligned} \tag{2.40}$$

then we can have $\widehat{\mathbb{H}}_1$ as **periodic boundary condition Hamiltonian** with $\hat{c}_{L+1} = \hat{c}_1$ and $\widehat{\mathbb{H}}_0$ is **anti-periodic boundary condition Hamiltonian** with $\hat{c}_{L+1} = -\hat{c}_1$. By solving them we can obtain the solution from the block form Hamiltonian like in the expression 2.39. Or we can write the $\widehat{\mathbb{H}}_{\mathrm{p}}$ even more compact:

$$\widehat{\mathbb{H}}_{\mathrm{p}=0,1} = -J \sum_{j=1}^{L} \left(\hat{c}_j^\dagger \hat{c}_{j+1} + \hat{c}_j^\dagger \hat{c}_{j+1}^\dagger + \text{ H.c. } \right) + h \sum_{j=1}^{L} \left(2\hat{n}_j - 1 \right), \tag{2.41}$$

with $\hat{c}_{L+1} = (-1)^{p+1} \hat{c}_1$, where $p = 1$ (odd parity) and $p = 0$ (even parity) represent for periodic and anti-periodic boundary conditions, respectively.

In order to diagonalise the Hamiltonian, firstly we go to a plane wave basis, which are:

$$\begin{cases} \hat{c}_k = \frac{1}{\sqrt{L}} \sum_j \hat{c}_j e^{-ikj}, \\ \hat{c}_j = \frac{1}{\sqrt{L}} \sum_k \hat{c}_k e^{ikj}. \end{cases} \tag{2.42}$$

$\widehat{\mathbb{H}}_{\mathrm{p}}$ written in plane wave basis is given:

$$\widehat{\mathbb{H}}_{\mathrm{p}} = -J \sum_{k}^{\mathcal{K}_{\mathrm{p}}} \left(2\cos k \hat{c}_k^{\dagger} \hat{c}_k + \left(e^{ik} \hat{c}_k^{\dagger} \hat{c}_{-k}^{\dagger} + \text{ H.c. } \right) \right) + h \sum_{k}^{\mathcal{K}_{\mathrm{p}}} \left(2\hat{c}_k^{\dagger} \hat{c}_k - 1 \right), \tag{2.43}$$

or we can rewrite:

$$\widehat{\mathbb{H}}_{\mathrm{p}} = \sum_{k}^{\mathcal{K}_{\mathrm{p}}} \left((h - J\cos k) \left(\hat{c}_k^{\dagger} \hat{c}_k - \hat{c}_{-k} \hat{c}_{-k}^{\dagger} \right) - J \left(e^{ik} \hat{c}_k^{\dagger} \hat{c}_{-k}^{\dagger} + \text{ H.c. } \right) \right), \tag{2.44}$$

where

$$\mathcal{K}_{\mathrm{p}=1} = \left\{ k = \frac{2n\pi}{L}, \text{ with } n = -\frac{L}{2} + 1, \cdots, 0, \cdots, \frac{L}{2} \right\}, \tag{2.45}$$

and

$$\mathcal{K}_{\mathrm{p}=0} = \left\{ k = \pm \frac{(2n-1)\pi}{L}, \text{ with } n = 1, \cdots, \frac{L}{2} \right\}. \tag{2.46}$$

It is useful if we can rewrite $\widehat{\mathbb{H}}_{\mathrm{p}}$ again in the form of $\widehat{\mathbb{H}}_{\mathrm{p}} = \sum_k \widehat{H}_k$ and consider only the positive value of k. The problem is in the $\mathrm{p} = 1$ case, $k = 0$ and $k = \pi$ do not have the $-k$ corresponding counterpart, so we have to write them out the other couples values of k, which means:

$$\widehat{\mathbb{H}}_0 = \sum_{k \in \mathcal{K}_0^+} \widehat{H}_k, \tag{2.47}$$

and

$$\widehat{\mathbb{H}}_1 = \sum_{k \in \mathcal{K}_1^+} \widehat{H}_k + \widehat{H}_{k=0,\pi}, \tag{2.48}$$

where:

$$\mathcal{K}_{\mathrm{p}=1}^+ = \left\{ k = \frac{2n\pi}{L}, \text{ with } n = 1, \cdots, \frac{L}{2} - 1 \right\}, \tag{2.49}$$

$$\mathcal{K}_{\mathrm{p}=0}^+ = \left\{ k = \frac{(2n-1)\pi}{L}, \text{ with } n = 1, \cdots, \frac{L}{2} \right\}, \tag{2.50}$$

$$\widehat{H}_{k=0,\pi} = -2J\left(\hat{n}_0 - \hat{n}_\pi\right) + 2h\left(\hat{n}_0 + \hat{n}_\pi - 2\right), \tag{2.51}$$

and

$$\widehat{H}_k = 2(h - J\cos k)\left(\hat{c}_k^{\dagger} \hat{c}_k - \hat{c}_{-k} \hat{c}_{-k}^{\dagger} \right) - 2J\sin k \left(i\hat{c}_k^{\dagger} \hat{c}_{-k}^{\dagger} - i\hat{c}_{-k} \hat{c}_k \right). \tag{2.52}$$

To diagonalize the Hamiltonian, we want to use the Bogoliubov transformation in which new fermion creation operators $\hat{\gamma}^\dagger$ are formed as linear combination of $\hat{c}_k^\dagger$ and $\hat{c}_k$ in order to remove the terms in Hamiltonian which do not conserve the particle number.

$\widehat{H}_k$ can be also written as:

$$\widehat{H}_k = \left(\hat{c}_k^\dagger, \hat{c}_{-k}\right) \begin{pmatrix} 2(h - J\cos k) & -2Ji\sin k \\ 2Ji\sin k & -2(h - J\cos k) \end{pmatrix} \begin{pmatrix} \hat{c}_k \\ \hat{c}_{-k}^\dagger \end{pmatrix} \tag{2.53}$$

Note that the vacuum of $\widehat{\mathbb{H}}$ now is not the vacuum of the operator $\hat{c}_k$ because of the presence of the off-diagonal term $\hat{c}_k^\dagger \hat{c}_k - \hat{c}_{-k}\hat{c}_{-k}^\dagger$ in the Hamiltonian. Then we expect to get the $\widehat{H}_k$ in the form

$$\widehat{H}_k = \sum_k \epsilon_k \hat{\gamma}_k^\dagger \hat{\gamma}_k + \text{const}, \tag{2.54}$$

so that the single particle excitations are identifiable above the vacuum state $\hat{\gamma}_k |0\rangle = 0$. It can be obtained by applying the canonical transformation (or Bogoliubov transformation):

$$\begin{pmatrix} \hat{\gamma}_k \\ \hat{\gamma}_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k^* & v_k^* \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \hat{c}_k \\ \hat{c}_{-k}^\dagger \end{pmatrix}, \tag{2.55}$$

And the inverse transformation is:

$$\begin{pmatrix} \hat{c}_k \\ \hat{c}_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & -v_k^* \\ v_k & u_k^* \end{pmatrix} \begin{pmatrix} \hat{\gamma}_k \\ \hat{\gamma}_{-k}^\dagger \end{pmatrix}. \tag{2.56}$$

The $\widehat{H}_k$ after rewritten is terms of the Bogoliubov transformation is:

$$\begin{aligned} \widehat{H}_k = \left(\hat{\gamma}_k^\dagger, \hat{\gamma}_{-k}\right) \begin{pmatrix} u_k^* & v_k^* \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} 2(h - J\cos k) & -2Ji\sin k \\ 2Ji\sin k & -2(h - J\cos k) \end{pmatrix} \\ \times \begin{pmatrix} u_k & -v_k^* \\ v_k & u_k^* \end{pmatrix} \begin{pmatrix} \hat{\gamma}_k \\ \hat{\gamma}_{-k}^\dagger \end{pmatrix} = \left(\hat{\gamma}_k^\dagger, \hat{\gamma}_{-k}\right) \begin{pmatrix} \epsilon_k & 0 \\ 0 & -\epsilon_k \end{pmatrix} \begin{pmatrix} \hat{\gamma}_k \\ \hat{\gamma}_{-k}^\dagger \end{pmatrix}, \end{aligned} \tag{2.57}$$

or we can write:

$$\widehat{H}_k = \epsilon_k(\hat{\gamma}_k^\dagger \hat{\gamma}_k + \hat{\gamma}_{-k}^\dagger \hat{\gamma}_{-k} - 1) \tag{2.58}$$

In order to recast $\widehat{H}_k$ in the diagonal form, we demand that $(u_k, v_k)^T$ is an eigenvector of the Hamiltonian matrix:

$$\begin{pmatrix} 2(h - J\cos k) & -2Ji\sin k \\ 2Ji\sin k & -2(h - J\cos k) \end{pmatrix} \begin{pmatrix} u_k \\ v_k \end{pmatrix} = -\epsilon_k \begin{pmatrix} u_k \\ v_k \end{pmatrix}. \tag{2.59}$$

This 2×2 eigenvalue problem is easily solved to yield the spectrum:

$$\epsilon_k = 2J\sqrt{\left(\cos k - \frac{h}{J}\right)^2 + \sin^2 k}. \tag{2.60}$$

From that, we find that the actual global ground state energy is the one in the $p = 0$ (even) sector:

$$E_0^{ABC} = -\sum_{k>0}^{ABC} \epsilon_k, \tag{2.61}$$

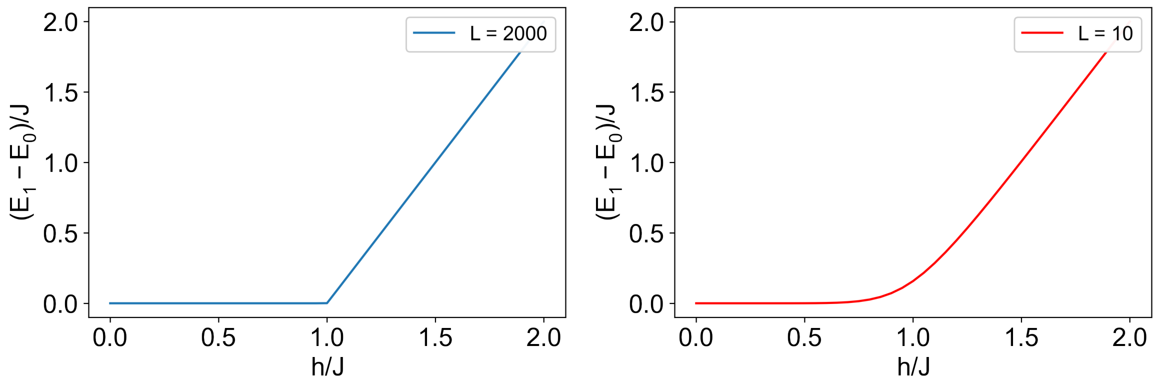
and the minimum energy of $p = 1$ (odd) sector is actually first excited state:

$$E_0^{PBC} = -2J - \sum_{0<k<\pi}^{PBC} \epsilon_k. \tag{2.62}$$

2.5 Exact diagonalization calculation of 1D TFIM

The diagonalization of the full Hamiltonian at (2.4) could be also performed numerically. This method which we use to numerically compute properties of a quantum system is called Exact Diagonalization (ED). Since eigenvectors and eigenvalues of matrices can be obtained up to high precision with modern algorithms, ED is a powerful and reliable numerical tool.

Figure below is the comparison between the analytical solution we got in the section 2.4 and the ED method's solution of the gap between first excited state and ground state energy. In the figure, the gap $E_1 - E_0$ is actually the energy splitting of $E_0^{PBC} - E_0^{ABC}$ which we just explained in the last section. We found that there is a two-fold degeneracy in the ferromagnetically ordered phase region $h < J$, that is why the energy splitting is zero. But in the quantum disordered regions $h > J$, the gap becomes finite with the value increasing with the strength of transverse field, $\Delta_1 = E_0^{PBC} - E_0^{ABC} = 2(h - J)$.



(a) Analytical method

(b) Exact Diagonalization method

Figure 2.1: Comparison between the analytical solution we got in the section 2.4 and the ED solution of the gap between first excited state and ground state energy.

Next, we want to observe the behavior of the order parameter, magnetization, in this model. The definition of the magnetization operator is given as:

$$\widehat{M} = \frac{1}{L}\sum_{j=1}^{L} \hat{\sigma}_j^x. \tag{2.63}$$

Unfortunately, there is no way to use Jordan-Wigner transformation to convert $\hat{\sigma}^x$ into spinless fermion, so we can only observe the order parameter in the ED method. However, even with ED method, we still can not observe the order parameter right now, because the average magnetization with the symmetrized ground state, which we mentioned in the expression (2.13), will always give us 0:

$$M = \left\langle 0_J^{\pm}\right| \widehat{M} \left|0_J^{\pm}\right\rangle = \left\langle 0_J^{\pm} \middle| 0_J^{\mp}\right\rangle = 0. \tag{2.64}$$

So the idea is to add a very small perturbation into the system to split the degenerate states. The Hamiltonian now becomes:

$$\widehat{H} = -J\sum_j \hat{\sigma}_j^x \hat{\sigma}_{j+1}^x - h\sum_j \hat{\sigma}_j^z + \epsilon_p \sum_{j=1} \hat{\sigma}_j^x, \tag{2.65}$$

where the last term is the small perturbation we would like to add into the system.

In Fig. 2.2, we show the change of ground state magnetization when we keep increas-

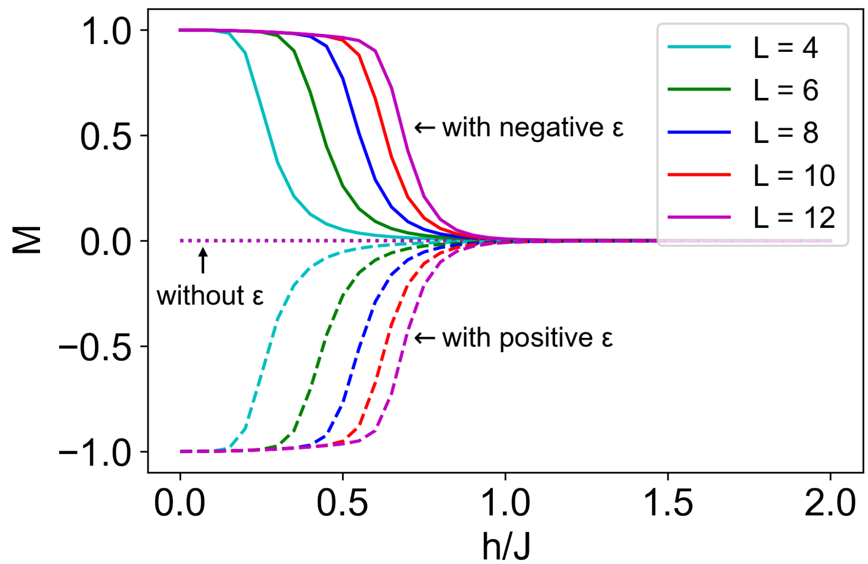


Figure 2.2: Behavior of ground state magnetization varies on the change of the magnitude of external field. The dotted line represents for the magnetization before adding the perturbation. The solid and the dashed line represent for the magnetization with negative and positive value of perturbation strength, respectively.

ing the external field. The dotted line represents for the magnetization before adding the perturbation. The solid and the dashed line represent for the magnetization with negative and positive value of perturbation strength ϵ_p, respectively. It is clear to see that by applying the small perturbation into the system, we successfully observe the behavior of order parameter magnetization to realize the presence of quantum phase transition. The ground state magnetization M is non-zero in the ordered phase regions, $h < J$, and zero in the disordered phase regions (above the critical point $h = J$) as we expected for the behaviors of a order parameter during phase transition.

Chapter 3

Boundary-frustrated transverse field Ising model

3.1 Introduction to boundary-frustration

In this chapter, we focus on the system with antiferromagnetic interactions in the Ising term of the transverse field Ising model and investigate under which circumstances the system is frustrated. Without loss of generality, we change the sign of J in the Hamiltonian (2.4) for the convenient purposes, so $J > 0$ represent for the antiferromagnetic interactions from now on. In order to understand frustration, the easiest way is to look at the classical Ising model, which is the prototypical model to visualize the classical frustration:

$$\widehat{H} = J \sum_{j=1}^{L} \hat{\sigma}_j^x \hat{\sigma}_{j+1}^x. \tag{3.1}$$

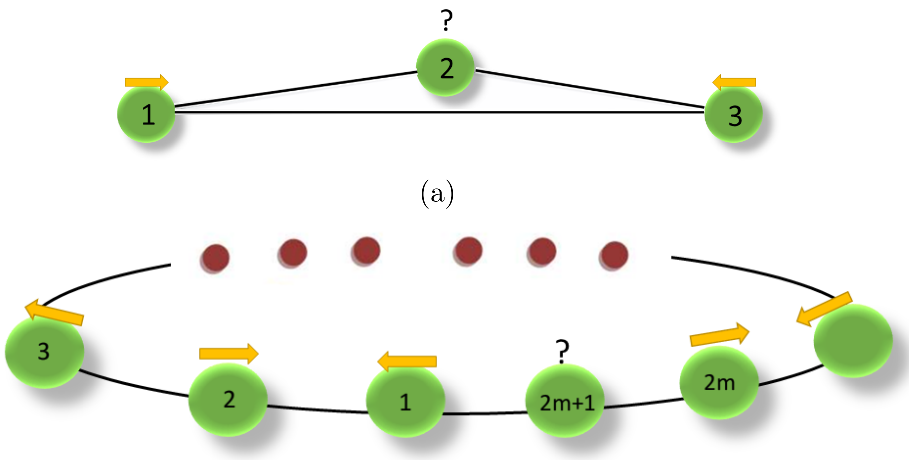
If the system size L is an even number, the situation is totally similar to the case of ferromagnetic interactions which we discussed in the last chapter. In this regime, all distinct interactions in the system can be minimized independently for the case of two Néel states, defined as:

$$|\rightarrow\leftarrow\rightarrow ...\rangle \,, \tag{3.2}$$

and

$$|\leftarrow\rightarrow\leftarrow ...\rangle \,. \tag{3.3}$$

However, with the odd system sizes $L = 2m + 1$, all the interactions cannot be minimized simultaneously (it can be seen in the Fig. 3.1) because one bond avoids the minimization and then lead to $2L$ degenerate ground states. This phenomenon is called frustration (or geometrical frustration).



(b)

Figure 3.1: The Ising ring with odd number of lattice sites: a) $L = 3$, b) $L = 2m+1$ ($m = 1, 2, 3, ...$), which exhibits the ring frustrations due to antiferromagnetic interaction ($J > 0$).

In this chapter, we want to use that concept of frustration to study the behaviors of the transverse field Ising chain in case of the antiferromagnetic ($J > 0$) system with periodic boundary condition and odd system sizes (L is odd). We choose the last bond in the chain to be the defected one, so that we can observe the model by changing the strength of that interaction. The Hamiltonian now is:

$$\widehat{H} = J \sum_{j}^{L-1} \hat{\sigma}_j^x \hat{\sigma}_{j+1}^x - h \sum_{j}^{L} \hat{\sigma}_j^z + J_{link} \hat{\sigma}_L^x \hat{\sigma}_1^x. \tag{3.4}$$

For this study, we want to examine the system properties with the larger system sizes, that would be time-consuming and computationally costly. In order to improve our ED simulation, we take the advantages of the sparse matrix and use the Lanczos algorithm.

3.2 Lanczos algorithm for ground state calculation

The ground state of the system is obtained via exact diagonalization using the Lanczos method. A key practical advantage of this approach is that it requires only matrix-vector multiplications rather than explicit storage of the full Hamiltonian matrix — only non-zero elements need to be retained, resulting in significant memory savings for large sparse systems.

The central idea of the Lanczos algorithm is to construct a Krylov subspace $\mathcal{K}$ by iteratively applying the Hamiltonian H to an initial random state $|\phi_0\rangle$:

$$\mathcal{K} = \left\{|\phi_0\rangle,\ H|\phi_0\rangle,\ H^2|\phi_0\rangle,\ \ldots,\ H^M|\phi_0\rangle\right\}. \tag{3.5}$$

Since M is typically chosen to be much smaller than the full Hilbert space dimension, $\mathcal{K}$ constitutes only a low-dimensional projection of the complete Hilbert space. An orthonormal basis of this subspace is then constructed through the three-term recurrence relation:

$$|\phi_{n+1}\rangle = H|\phi_n\rangle - a_n|\phi_n\rangle - b_n^2|\phi_{n-1}\rangle, \tag{3.6}$$

where the coefficients a_n and b_n^2 are defined as

$$a_n = \frac{\langle\phi_n|H|\phi_n\rangle}{\langle\phi_n|\phi_n\rangle}, \qquad b_n^2 = \frac{\langle\phi_n|\phi_n\rangle}{\langle\phi_{n-1}|\phi_{n-1}\rangle}, \tag{3.7}$$

with the boundary conditions $b_0 = 0$ and $|\phi_{-1}\rangle = 0$. Crucially, at each step only three vectors $\{|\phi_{n+1}\rangle, |\phi_n\rangle, |\phi_{n-1}\rangle\}$ must be held in memory simultaneously, keeping the memory footprint minimal.

A particularly elegant consequence of this construction is that, in the normalized basis $|n\rangle \equiv |\phi_n\rangle/\sqrt{\langle\phi_n|\phi_n\rangle}$, the projected Hamiltonian takes a tridiagonal form:

$$H = \begin{pmatrix} a_0 & b_1 & 0 & 0 & \cdots & 0 \\ b_1 & a_1 & b_2 & 0 & \cdots & 0 \\ 0 & b_2 & a_2 & b_3 & \cdots & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & 0 & 0 & a_N \end{pmatrix}, \tag{3.8}$$

which can be diagonalized efficiently by standard numerical methods. The iteration is terminated once the lowest eigenvalue converges to within a prescribed tolerance, typically requiring $N_{\text{step}} \approx 10$–$200$ iterations depending on the system size. The rate of convergence is governed by the spectral gap: a smaller gap between the two lowest eigenvalues leads to slower convergence. In the case of a degenerate ground state, the algorithm converges to an arbitrary vector within the degenerate subspace.

Once the tridiagonal eigenvalue problem is solved, the ground state wavefunction $|\psi\rangle$ is first expressed in the Lanczos basis $\{|\phi_n\rangle\}$. Recovering its representation in the original Hilbert space basis requires a second Lanczos run, during which the expansion coefficients $\langle\psi|\phi_n\rangle$ are extracted at each iteration.

3.3 Frustration in antiferromagnetic ground state

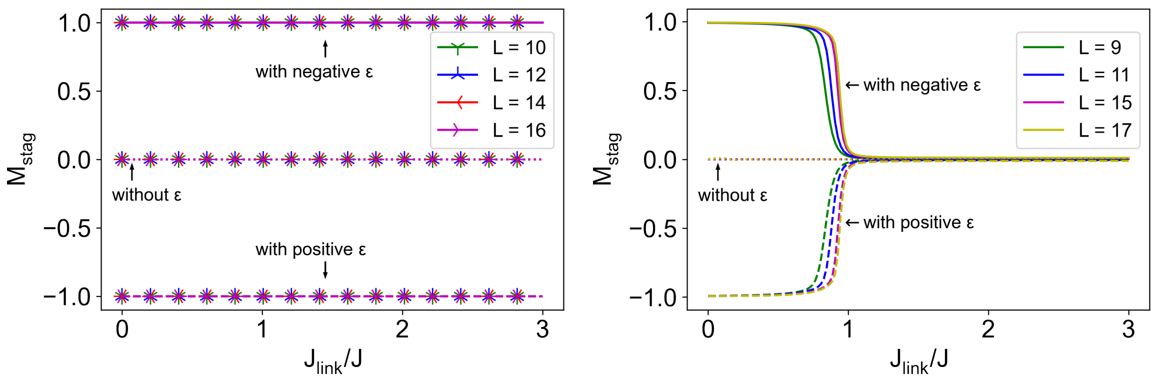
As in the case of ferromagnetic interactions of Ising model in the last chapter, we also want to examine the order parameter to see if there is actually any phase transition. The order parameter in this system is not the magnetization which we introduced earlier, but the staggered magnetization $\widehat{M}_{stag}$,

$$\widehat{M}_{stag} = \frac{1}{L}\sum_{j=1}^{L}(-1)^j \hat{\sigma}^x_j. \tag{3.9}$$

The staggered magnetization is numerically calculated and plotted in the Fig. 3.2 below for the values of the transverse field h/J is 0.2. We plot the staggered magnetization as a function of the defected bond strength J_{link} for different system sizes (even system sizes on the left and odd system sizes on the right). We use the same idea in the last chapter, which is to add a small perturbation into the Hamiltonian to split the degenerate states in case they exist. The perturbation, however, is also in the staggered form now. The Hamiltonian becomes:

$$\widehat{H} = J\sum_{j=1}^{L-1}\hat{\sigma}^x_j\hat{\sigma}^x_{j+1} - h\sum_{j}^{L}\hat{\sigma}^z_j + J_{link}\hat{\sigma}^x_L\hat{\sigma}^x_1 + \epsilon_p\sum_{j=1}^{L}(-1)^j\hat{\sigma}^x_j \tag{3.10}$$

Note that in the figures, the dotted line is the staggered magnetization before adding



(a) Even system size (b) Odd system size

Figure 3.2: Staggered magnetization of the boundary-frustrated transverse field Ising model as a function of defected bond strength for several values of the size L where a) even system size and b) odd system size, when the transverse field $h/J = 0.2$. The dotted line is the staggered magnetization before adding the small perturbation, the solid and the dashed line are the ones when we choose the value of ϵ_p is negative and positive, respectively.

the small perturbation, the solid and the dashed line are the ones when we choose the value of ϵ_p is negative and positive, respectively. We found that there is no phase transition for the even system sizes as there is no frustration. So the staggered magnetization is either $+1$, -1 or 0 depending on the ϵ_p value. But in the odd system size cases, as one increases J_{link} from inside the Néel phase, the staggered magnetization decreases and vanishes at the critical point (when $J_{link}/J = 1$). The plots are actually in agreement with our expectations. Because on the even chains when $h/J = 0.2$ which is small compared to $J = 1$, the two Néel states minimize all the local interactions, no matter what value of J_{link} is, thus the staggered magnetization remains unchanged during the change of J_{link}. For the odd Ising chains, when J_{link} is small, the system acts like a normal open boundary condition transverse field Ising model. But as J_{link} increases, the system starts suffering the frustration phenomenon. This motivates the phase transition to happen.

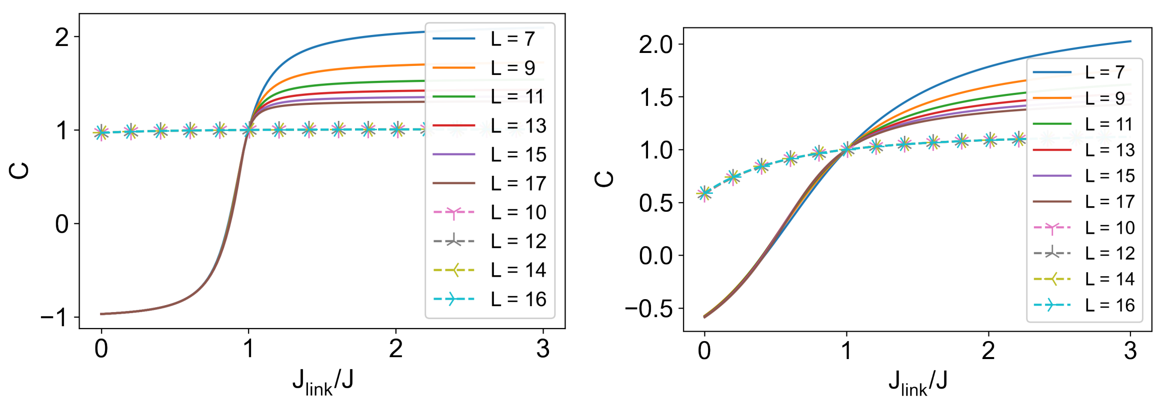
3.4 Correlation functions with frustration

The next quantity we concern is the longitudinal correlation function in the ground state,

$$C^x = \langle \hat{\sigma}_i^x \hat{\sigma}_j^x \rangle = \langle GS | \hat{\sigma}_i^x \hat{\sigma}_j^x | GS \rangle . \tag{3.11}$$

In order to study the change of the correlation between the last and the first spins, we will want to compare that correlation with the one between two spins in the middle of the chain. That means we change the value of J_{link} and examine how the quantity C changes, where C is given as:

$$C = \frac{C_1^x}{C_2^x} = \frac{\langle \hat{\sigma}_L^x \hat{\sigma}_1^x \rangle}{\langle \hat{\sigma}_{\frac{L}{2}}^x \hat{\sigma}_{\frac{L}{2}+1}^x \rangle}. \tag{3.12}$$



(a) $h/J = 0.2$

(b) $h/J = 0.7$

Figure 3.3: The correlation function of the defected bond divided by the bond in the middle of the chain, during the increase of J_{link} with a) $h/J = 0.2$ and b) $h/J = 0.7$, with different system size.

In the Fig. 3.3, we show the change of quatity C with the increase of J_{link} with different system size, where C is the correlation function of the defected bond divided by the bond in the middle of the chain. The odd system size cases are plotted in the solid lines, while the even ones cases are plotted in the dashed lines with different symbols.

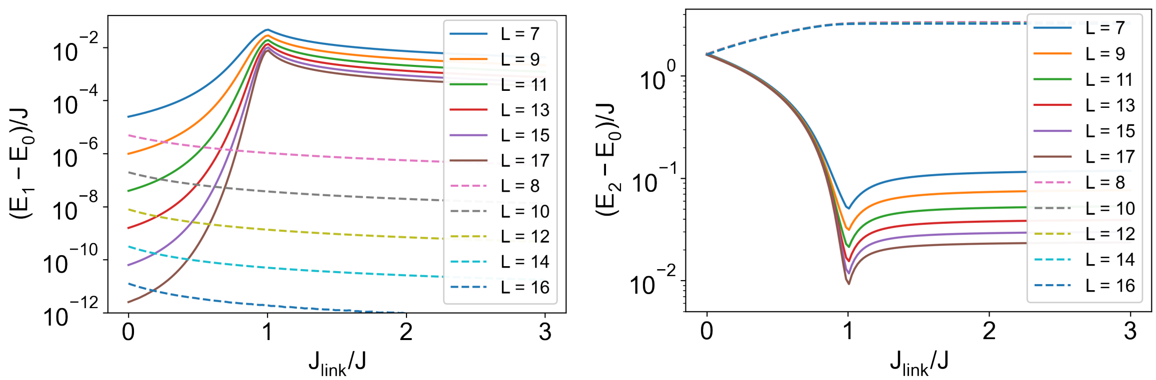
For the even system size cases, there is no frustration phenomenon, so C is always around 1 in the $h/J = 0.2$ and changed a bit in the $h/J = 0.7$ case. For the odd system sizes, firstly the chain acts like the open boundary chain for small J_{link}, so that the first and the last spins tend to have the same direction while the spins in the middle are in opposite directions. When $J_{link} > J$, however, two spins in the defected bond now become to point in the opposite directions. This explains why C is negative when $J_{link} < J$ and then increases with J_{link} increasing, finally saturating to a positive value, $C > 1$ at large J_{link}.

3.5 Energy gaps in the boundary-frustrated tranverse field Ising model

Finally, we want to observe the energy differences of the system in both $h/J = 0.2$ and $h/J = 0.7$ cases. We call the energy difference between the first excited level and ground state energy is first gap and the second excited level and ground state energy is second gap for short.

What we can see in the Fig. 3.4a is that when the value of J_{link}/J is 0, which means the system goes deeply into the antiferromagnetic phase, the ground state degeneracy is exact for large enough system-size just like the system in the open boundary condition. The first gaps start decreasing exponentially with increasing the system size L when $J_{link}/J < 1$ for both odd and even system sizes (see Fig. 3.5a for the odd sizes when $J_{link}/J = 0.5$): $E_1 - E_0 = c_1 e^{-\alpha L}$. For the even system sizes, however, the first gaps always decrease exponentially with the increase of the size, because the frustration does not appear in these cases. On the contrary, for the odd system sizes, what we see is that there is a cusp in the first energy gap when the value of J_{link}/J at exactly 1. This is the signal of phase transition. And after that, the first gap decays no longer exponentially with the system sizes, but decreases instead with a power of the size (see the Fig. 3.5b for $J_{link}/J = 2$): $E_1 - E_0 = c_2 L^{-\beta}$. The cusp in the first gap can be also seen in the Fig. 3.6a for the $h/J = 0.7$ case.

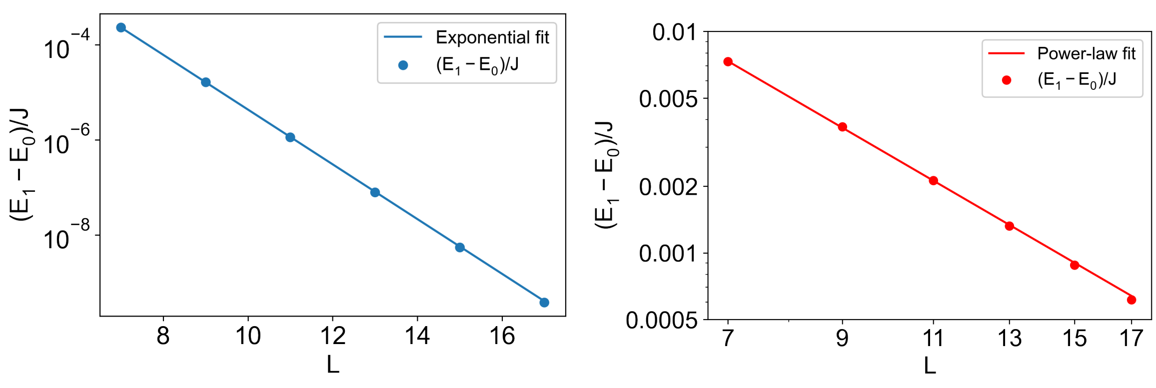
We can also observe the phase transition in the second gap (see Fig. 3.4b for $h/J = 0.2$). For the even system sizes, the gaps are actually the spectral gaps between the ground state and the spectral lines. They are always non-zero and saturate to a constant value. In the odd system sizes, the different point is that there is a cusp at $J_{link}/J = 1$ which is the signal of phase transition, and after that the second gaps decay obeying power-law in the size. The same happen also for $h/J = 0.7$ in the Fig. 3.6b, but since the system is very closed to the paramagnetic phase and the original Ising phase transition, the finite size effect is much more prominent.



(a)

(b)

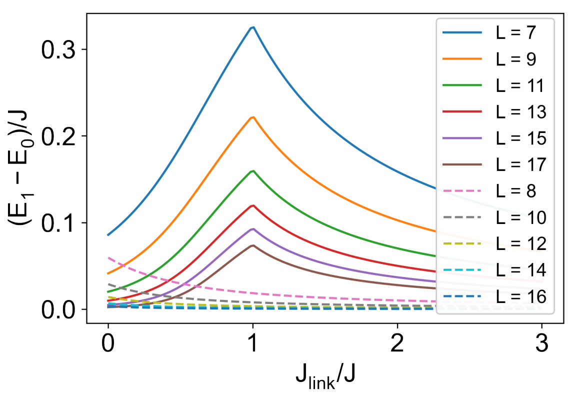
Figure 3.4: Energy differences between the energy of the first (left), second (right) excited level and the ground state with the change of J_{link} at $h/J = 0.2$ of the different system sizes.



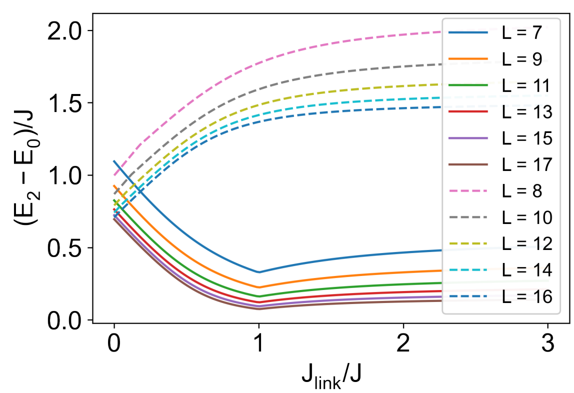
(a)

(b)

Figure 3.5: The curve fitting of first energy gap for odd system sizes in case of $h = 0.2$ with a) exponential fit at the point $J_{link}/J = 0.5$ and b) power-law fit at the point $J_{link}/J = 2$.



(a)



(b)

Figure 3.6: Energy differences between the energy of the first (left), second (right) excited level and the ground state with the change of J_{link} at $h/J = 0.7$ of the different system sizes.

Chapter 4

Conclusions

In this thesis, we showed the corresponding between a spin chain and a 1D fermionic system. This correspondence is then used it to study the quantum phase transition and geometrical frustration in the transverse field Ising model. In the transverse field Ising model with the ferromagnetic interactions between spins, we studied the quantum phase transition by observing the behaviors of the order parameter and the energy differences for both analytical and numerical methods. We found that there is a two-fold degeneracy in the ferromagnetically ordered phase region $h < J$ and finite gaps with the value increasing with the strength of transverse field in the quantum disordered regions $h > J$. The order parameter, ground state magnetization M is non-zero in the ordered phase regions, $h < J$, and zero in the disordered phase regions (above the critical point $h = J$) as we expected for the behaviors of a order parameter during phase transition. The geometrical frustration is happened in the transverse field Ising model with the antiferromagnetic interactions between spins, when the number of spins (or the number of system size in the fermionic system) is odd with periodic boundary condition. This frustration also motivates the quantum phase transition. We studied this interesting phenomenon by considering the transverse field Ising model in the presence of a bond defect called J_{link}. By doing so, we can realize the phase transition when we look at the cusp in the energy splitting and the order parameter of the system (staggered magnetization). The correlation between two spins in the defected bond is also examined to see the behaviors of them during the frustration. It is remarkable that although numerical calculations have been performed in small chains, the Quantum Phase Transition can be still revealed, which in principle can be seen only in the thermodynamic limit. All of the codes we used in this thesis can be found in my github page [5].

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Appendix A: Pauli matrices

The definition of Pauli matrices are:

$$\hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}^y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \text{ and } \quad \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{A.1}$$

The ladder operators are derived from the Pauli matrices:

$$\hat{\sigma}^+ = \frac{1}{2}\left(\hat{\sigma}^x + \mathrm{i}\hat{\sigma}^y\right) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{A.2}$$

and

$$\hat{\sigma}^- = \frac{1}{2}\left(\hat{\sigma}^x - \mathrm{i}\hat{\sigma}^y\right) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{A.3}$$

in which:

$$\hat{\sigma}^+|\downarrow\rangle = |\uparrow\rangle, \quad \hat{\sigma}^+|\uparrow\rangle = 0, \quad \hat{\sigma}^-|\uparrow\rangle = |\downarrow\rangle, \text{ and } \quad \hat{\sigma}^-|\downarrow\rangle = 0. \tag{A.4}$$

Commutation and anticommutation of Pauli matrices:

$$\left[\hat{\sigma}^i, \hat{\sigma}^j\right] = 2i\varepsilon_{ijk}\hat{\sigma}^k, \tag{A.5}$$

and

$$\left\{\hat{\sigma}^j, \hat{\sigma}^k\right\} = 2\delta_{jk}\hat{I}. \tag{A.6}$$

Appendix B: Bogoliubov transformation

Bogoliubov transformation is very effective for solving a Hamiltonian which has the quadratic form:

$$\widehat{H} = \sum_{ij} \hat{c}_i^\dagger A_{ij} \hat{c}_j + \frac{1}{2} \sum_{ij} \hat{c}_i^\dagger B_{ij} \hat{c}_j^\dagger + \text{H.c.} \tag{B.1}$$

We expect to transform this kind of complicated Hamiltonian into the friendly one:

$$\widehat{H} = \sum_k \epsilon_k \hat{d}_k^\dagger \hat{d}_k + \text{const}, \tag{B.2}$$

Firstly, we perform $\hat{d}_k$ and $\hat{d}_k^\dagger$ in the form of linear transformation of $\hat{c}_i$ and $\hat{c}_i^\dagger$:

$$\hat{d}_k = \sum_i (m_{ki}\hat{c}_i + n_{ki}\hat{c}_i^\dagger), \tag{B.3}$$

$$\hat{d}_k^\dagger = \sum_i (m_{ki}\hat{c}_i^\dagger + n_{ki}\hat{c}_i). \tag{B.4}$$

With the condition of getting the $\hat{d}_k$ which satisfy the fermionic anticommutation relations, we have:

$$\sum_i (m_{ki}m_{k'i} + n_{ki}n_{k'i}) = \delta_{kk'}, \tag{B.5}$$

$$\sum_i (m_{ki}n_{k'i} - m_{k'i}n_{ki}) = 0. \tag{B.6}$$

And from (B.2), we also have:

$$\{\hat{d}_k, \widehat{H}\} = \epsilon_k \hat{d}_k \tag{B.7}$$

Subtituting (B.3) and (B.4) into (B.7),

$$\epsilon_k m_{ki} = \sum_j (m_{kj}A_{ji} - n_{kj}B_{ji}), \tag{B.8}$$

$$\epsilon_k n_{qi} = \sum_j (m_{kj} B_{ji} - n_{kj} A_{ji}). \tag{B.9}$$

Or we can also rewrite:

$$\phi_k (A - B) = \epsilon_k \psi_k, \tag{B.10}$$

$$\psi_k (A + B) = \epsilon_k \phi_k. \tag{B.11}$$

where:

$$(\phi_k)_i = m_{ki} + n_{ki}, \tag{B.12}$$

$$(\psi_k)_i = m_{ki} - n_{ki}. \tag{B.13}$$

From that we have:

$$(A + B)(A - B) = \phi_k = \epsilon_k^2 \phi_k, \tag{B.14}$$

$$\psi_k (A + B)(A - B) = \epsilon_k^2 \psi_k. \tag{B.15}$$

So the $2N \times 2N$ matrix diagonalization problem is now reduced to the $N \times N$ eigenvalue problem. In order to get the const value in the (B.2), we match the trace of $\widehat{H}$ in the (B.1) and the one in the (B.2). The value of constant we get is:

$$\text{const} = \frac{1}{2} \left(\sum_i A_{ii} - \sum_k \epsilon_k \right). \tag{B.16}$$

Thus, the Hamiltonian after the Bogoliubov transformation is:

$$\widehat{H} = \sum_k \epsilon_k \hat{d}_k^\dagger \hat{d}_k + \frac{1}{2} \left(\sum_i A_{ii} - \sum_k \epsilon_k \right). \tag{B.17}$$